

Ehsan Aghazadeh

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Education

University of Massachusetts, Amherst Feb. 2023 – Feb. 2027 (Expected)
Masters/Ph.D. in Computer Science Massachusetts, Amherst
Advisors: Hossein Pishro-Nik, Hedyeh Beyhaghi Interests: Natural Language Processing, ML Theory, Deep Learning

University of Tehran Sep. 2017 – Aug. 2022
B.Sc. in Computer Engineering Tehran, Iran
GPA: 17.04/20

Publications

- On the Reliability of Networks of AI Agents: Density Evolution, Stopping Sets, and Architecture Optimization.** E. Aghazadeh, H. Pishro-Nik. *Ongoing*
- Ergodic Trajectory Design by Learned Pushforward Maps: Provable Coverage via Conditional Flow Matching.** E. Aghazadeh, M. Malekzadeh, A. Ghasemi, H. Pishro-Nik. *In Submission @ NeurIPS 2026*
- CGES: Confidence-Guided Early Stopping for Efficient and Accurate Self-Consistency.** E. Aghazadeh, A. Ghasemi, H. Beyhaghi, H. Pishro-Nik. *Efficient Reasoning @ NeurIPS 2025 (Spotlight) · In Submission @ EMNLP 2026*
- A Comprehensive Analysis for Visual Object Hallucination in Large Vision-Language Models.** L. Jing, H. Chen, E. Aghazadeh, X. E. Wang, X. Du. *KnowledgeFM @ ACL 2025*
- From RAGs to Rich Parameters: Probing how Language Models Utilize External Knowledge over Parametric Information for Factual Queries.** H. Wadhwa, R. Seetharaman, S. Aggarwal, R. Ghosh, S. Basu, S. Srinivasan, W. Zhao, S. Chaudhari, E. Aghazadeh. *BlackboxNLP @ EMNLP 2024*
- DecompX: Explaining Transformers Decisions by Propagating Token Decomposition.** A. Modaressi, M. Fayyaz, E. Aghazadeh, Y. Yaghoobzadeh, M. T. Pilehvar. *ACL 2023*
- BERT on a Data Diet: Finding Important Examples by Gradient-Based Pruning.** M. Fayyaz*, E. Aghazadeh*, A. Modaressi*, Y. Yaghoobzadeh, M. T. Pilehvar, S. E. Kahou. *ENLSP @ NeurIPS 2022* (*equal contribution)
- Metaphors in Pre-Trained Language Models: Probing and Generalization Across Datasets and Languages.** E. Aghazadeh*, M. Fayyaz*, Y. Yaghoobzadeh. *ACL 2022* (*equal contribution)
- Not All Models Localize Linguistic Knowledge in the Same Place: A Layer-wise Probing on BERToids' Representations.** M. Fayyaz*, E. Aghazadeh*, A. Modaressi, H. Mohebbi, M. T. Pilehvar. *BlackboxNLP @ EMNLP 2021* (*equal contribution)

Experience

Graduate Research Assistant March 2024 – Present
University of Massachusetts Amherst Amherst, MA

(multi-agent systems) (agentic AI) (test-time scaling) (self-consistency) (flow matching) (VLM)

- First to model **multi-agent AI systems** as factor graphs with **LDPC-inspired density evolution**, proving a theorem predicting the asymptotic fraction of unresolved subclaims as a function of agent accuracy and network topology; revealed sharp phase transitions and architecture optimization principles.
- Proposed the epushforward framework for ergodic trajectory design via **optimal-transport conditional flow matching**, proving an $O(1/\sqrt{K})$ ergodic convergence rate; enabled zero-shot deployment across multi-agent fleets without retraining.
- Proposed CGES, a Bayesian confidence-based framework achieving **25–40% reduction in inference cost** on GSM8K, MATH, MMLU, and GPQA with no accuracy loss; proved it converges to the correct answer matching majority voting with **fewer LLM calls**.
- Systematically probed VLMs to localize hallucination sources; showed **hallucinations predominantly stem from the vision encoder and projector**, offering targeted guidance for mitigation.

Undergraduate Research Assistant Jan. 2021 – Jan. 2023
University of Tehran & Tehran Institute for Advanced Studies Tehran, Iran

(mechanistic interpretability) (attribution patching) (probing) (emergent abilities)

- Proposed **DecompX**, a vector-based attribution method propagating decomposed token representations across all Transformer layers without mixing, **consistently outperforming** prior attribution methods on faithfulness benchmarks. (*ACL 2023*)
- First to probe PLMs for metaphorical knowledge; demonstrated **contextual representations encode metaphor in middle layers** with transferability across English, Spanish, Russian, and Farsi. (*ACL 2022*)

- Identified **weight-mixing evaluation in edge probing produces unreliable conclusions** due to norm disparities; showed linguistic knowledge localization varies substantially across BERT-family architectures. (*EMNLP 2021*)

Remote Research Assistant

Feb. 2022 – Nov. 2022

École de technologie supérieure

Montreal, Canada

data pruning sparsity distillation

- First to adapt **GraNd** and **EL2N** gradient-based importance metrics to NLP, enabling data pruning for fine-tuned language models; found early-phase scores unreliable in NLP unlike in vision.
- Showed pruning high-scoring examples **matches or surpasses** full-dataset fine-tuning performance, revealing significant redundancy in NLP training sets. (*NeurIPS 2022*)

Awards & Honors

- **Admitted with Grant**, Machine Learning Summer School (MLSS) NYC 2026, Columbia University. 2026
- **OpenAI API Researcher Access Program**, \$1,000 API credit grant for research use. 2025
- **Ranked 226th out of 150,000** in the Iranian National University Entrance Exam (Konkour). 2017

Academic Service

Program Committee / Reviewer for 20+ papers: EACL 2024, NAACL 2024, ACL 2024, ACL Rolling Review (since 2023)

Extra Curricular & Leadership

Member of Scientific Student Chapter

Jun. 2019 – Aug. 2020

University of Tehran, School of Electrical and Computer Engineering

Tehran, Iran

- Organized high-quality schools for Data Science, IoT, and Signal Processing; led 20+ events and headed the “Data Science Winter School” executive team.